

Basic Information

Experiment No.: 06

Date: 24-09-2026

Title: Java Map Interface using HashMap

Objective / Aim

To implement the Map interface using "HashMap" and demonstrate operations such as adding, accessing, searching, removing, and clearing key-value pairs.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

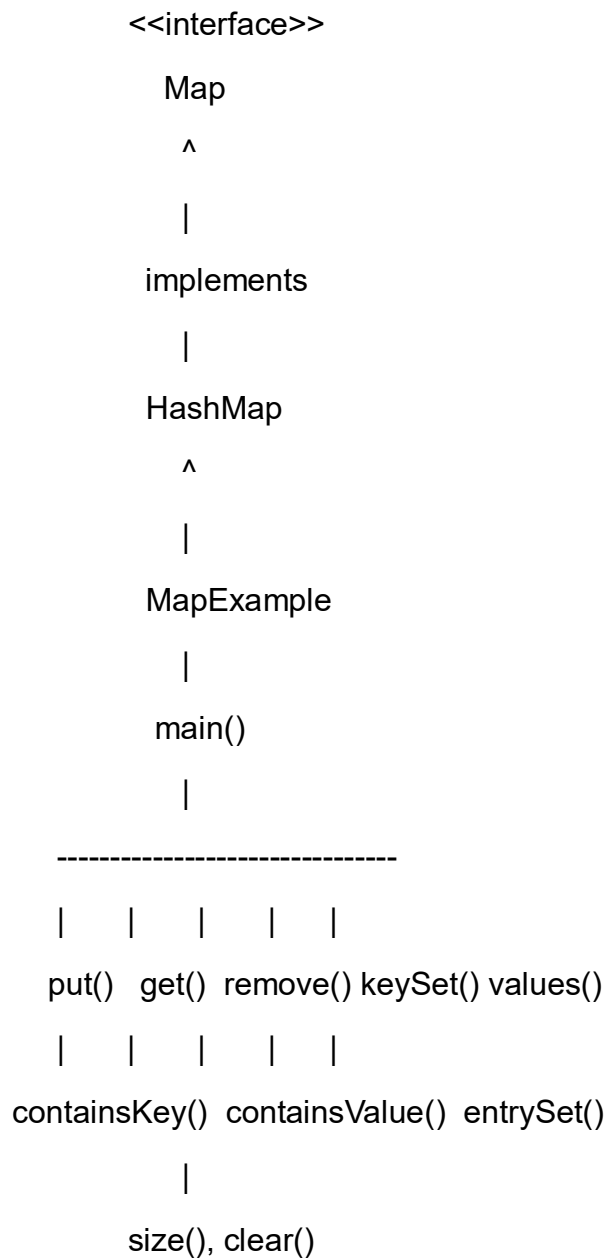
- Java JDK
 - Any Java IDE or Command Prompt
 - Operating System: Windows/Linux/macOS
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Algorithm / Logic Description

1. Create a "Map" using "HashMap".
2. Add key-value pairs using "put()".
3. Access a value using "get()".
4. Check keys and values using "containsKey()" and "containsValue()".
5. Remove an entry using "remove()".
6. Display keys using "keySet()".
7. Display values using "values()".
8. Display key-value pairs using "entrySet()".

9. Check size and empty status using "size()" and "isEmpty()".
 10. Remove all entries using "clear()".
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UML Class Diagram



Source Code

```
package collectionframework;
```

```
import java.util.Map;
```

```
import java.util.HashMap;
```

```
public class MapExample {
```

```
    public static void main(String[] args) {
```

```
        Map<Integer, String> map = new HashMap<>();
```

```
        // put()
```

```
        map.put(101, "Rahul");
```

```
        map.put(102, "Priya");
```

```
        map.put(103, "John");
```

```
        System.out.println("Map: " + map);
```

```
        // get()
```

```
        System.out.println("Value of key 102: " + map.get(102));
```

```
        // containsKey()
```

```
        System.out.println("Contains key 101: " +
```

```
            map.containsKey(101));
```

```
        // containsValue()
```

```
        System.out.println("Contains value John: " +
```

```
            map.containsValue("John"));
```

```
        // remove()
```

```
map.remove(103);

System.out.println("After remove: " + map);


// keySet()
System.out.println("Keys: " + map.keySet());


// values()
System.out.println("Values: " + map.values());


// entrySet()
System.out.println("Entries: " + map.entrySet());


// size()
System.out.println("Size: " + map.size());


// isEmpty()
System.out.println("Is Empty: " + map.isEmpty());


// clear()
map.clear();

System.out.println("After clear: " + map);
}
}
```

Compilation & Execution Commands

Save the file as:

MapExample.java

Since the program contains:

package collectionframework;

the file should be inside the "collectionframework" folder.

Compile:

javac MapExample.java

Run:

java collectionframework.MapExample

Output Screen

Note: "HashMap" does not guarantee insertion order, so the order of entries, keys, and values may vary.

Map: {101=Rahul, 102=Priya, 103=John}

Value of key 102: Priya

Contains key 101: true

Contains value John: true

After remove: {101=Rahul, 102=Priya}

Keys: [101, 102]

Values: [Rahul, Priya]

Entries: [101=Rahul, 102=Priya]

Size: 2

Is Empty: false

After clear: {}

Result

The Map interface was successfully implemented using "HashMap", and operations such as "put()", "get()", "containsKey()", "containsValue()", "remove()", "keySet()", "values()", "entrySet()", "size()", "isEmpty()", and "clear()" were successfully demonstrated.